CS6290 — Individual Evidence Pack (Milestone 3)

Role: Developer / Tooling (RPC Privacy Measurement Harness)

Name: Rongke Xiao (GitHub: Kstechero)
Milestone: 3
Repository: https://github.com/Kstechero/wallet-rpc-privacy-measurement
Role selected in group: Developer / Tool Developer

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1) What I contributed since the previous milestone

- Completed the project pipeline by extending the M2 experiment framework into a more complete **end-to-end measurement toolkit**: batch execution, structured summarization, and automated result visualization.
- Improved the batch experiment workflow so that repeated runs produce a **clean summary output** rather than accidentally appending stale rows from previous runs.
- Expanded the summarization layer to generate a richer consolidated CSV (`results/summary.csv`) with fields suitable for final analysis and reporting, including provider metadata, scenario metadata, latency statistics, and address-exposure indicators.
- Added an automated plotting step (`src/plots.py`) that converts the consolidated summary into final comparison figures for provider A/B evaluation across scenarios.
- Preserved the original M2 experiment matrix (`configs/m2/*.yaml`) to maintain continuity and comparability while improving the analysis and delivery pipeline around it.

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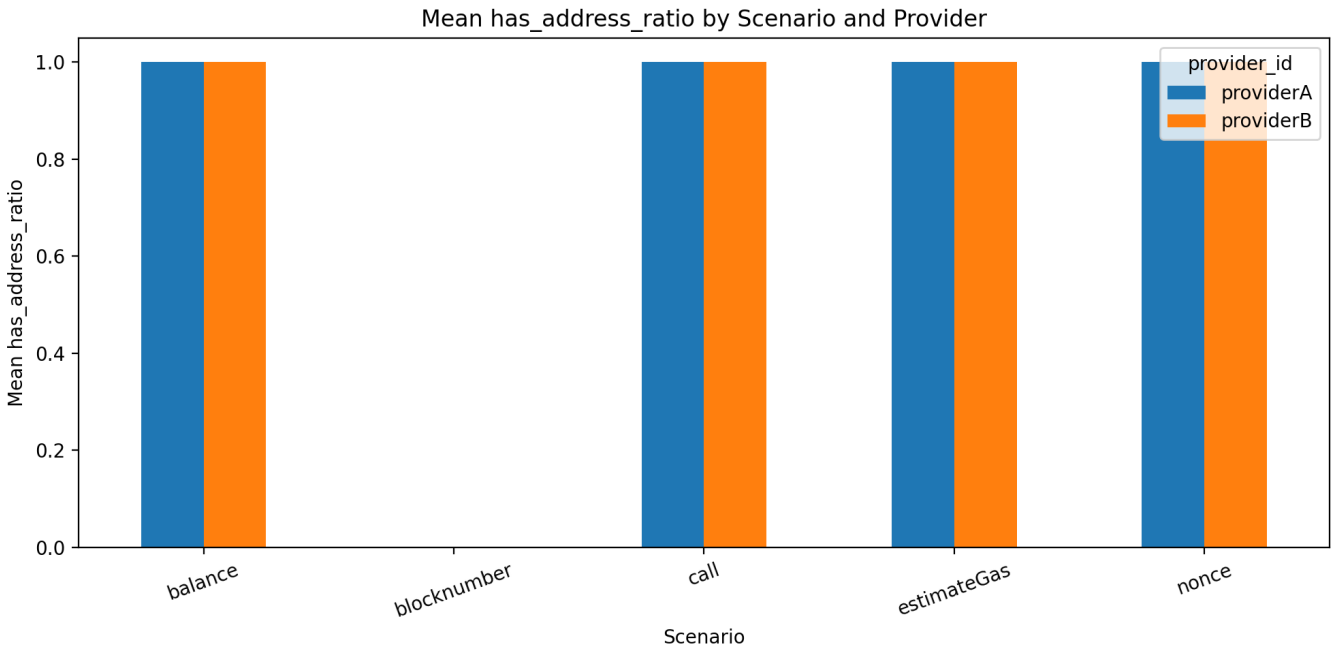
2) Evidence

#	Evidence type	Link / Reference	What this shows
0	repository	https://github.com/Kstechero/wallet-rpc-privacy-measurement	Public reference to the final project code, experiment configs, result summaries, and documentation.
1	code	`src/runner.py`, `src/rpc_client.py`, `src/scenarios.py`, `src/logger.py`	Core request-generation and structured logging pipeline for JSON-RPC measurements.
2	code	`src/batch_run.py`	End-to-end batch execution over the experiment matrix and generation of a fresh consolidated CSV summary.
3	code	`src/summarize.py`	Final summarization logic producing per-run metrics used for comparison and plotting.

#	Evidence type	Link / Reference	What this shows
4	code	`src/plots.py`	Automated generation of final provider-comparison figures from `results/summary.csv`.
5	config	`configs/m2/`	The experiment matrix used for final runs (provider A/B × scenario × interval × replicate).
6	result	`results/summary.csv`	Consolidated final summary used as the main analysis table for Milestone 3.
7	figure	`results/figures/latency_median.png`	Median latency comparison across scenarios and providers.
8	figure	`results/figures/latency_p95.png`	Tail-latency comparison across scenarios and providers.
9	figure	`results/figures/has_address_ratio.png`	Address-bearing request ratio comparison across scenarios and providers.
10	figure	`results/figures/ok_rate.png`	Availability comparison across scenarios and providers.

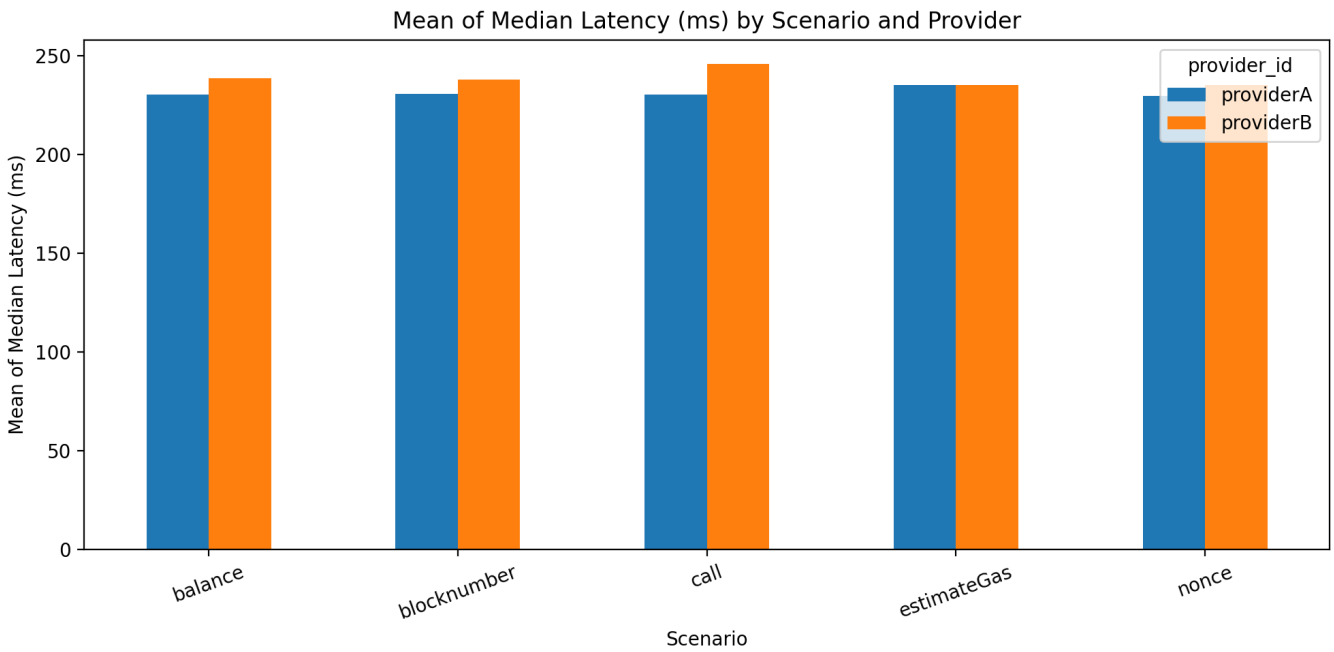
Screenshot evidence

Screenshot 1 — Final address-exposure comparison
What this shows: the figure `results/figures/has_address_ratio.png` verifies that the address-bearing proxy behaves as intended. `blocknumber` remains address-free (`has_address_ratio = 0`), while the other configured scenarios are detected as address-bearing under the current scenario definitions.



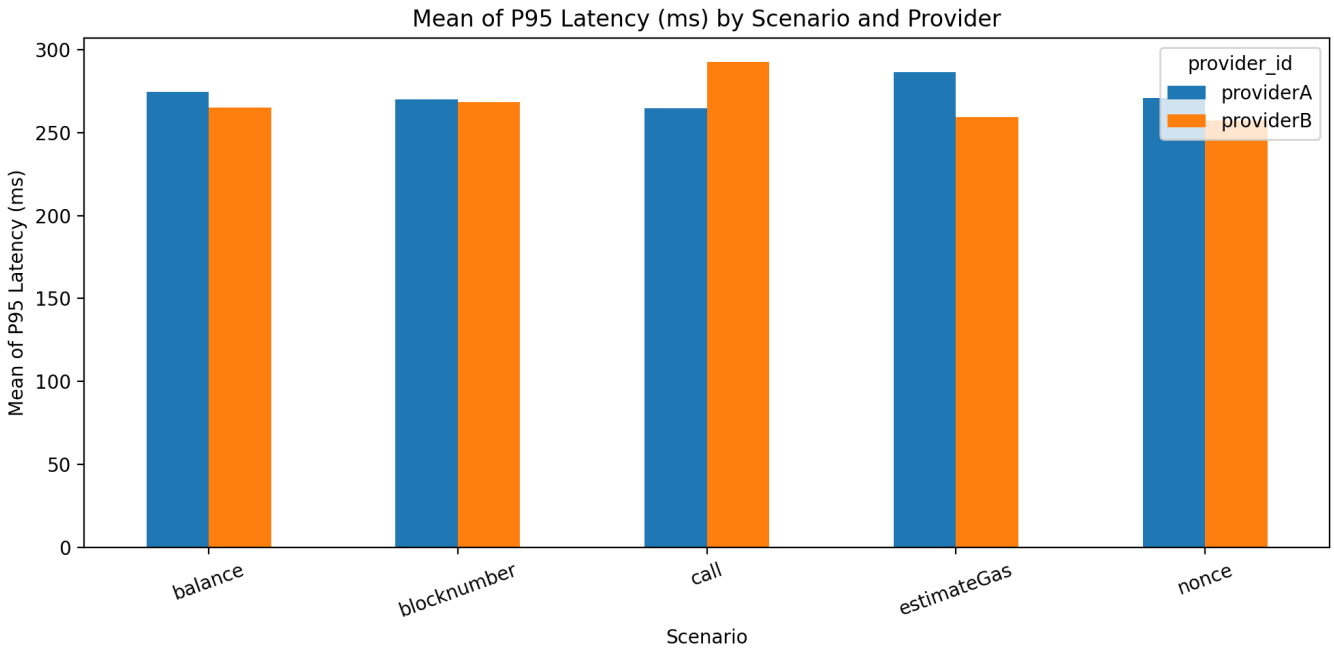
Screenshot 2 — Final median latency comparison

What this shows: the figure `results/figures/latency_median.png` provides a provider A/B comparison of median latency by scenario, allowing final performance observations beyond raw logs.



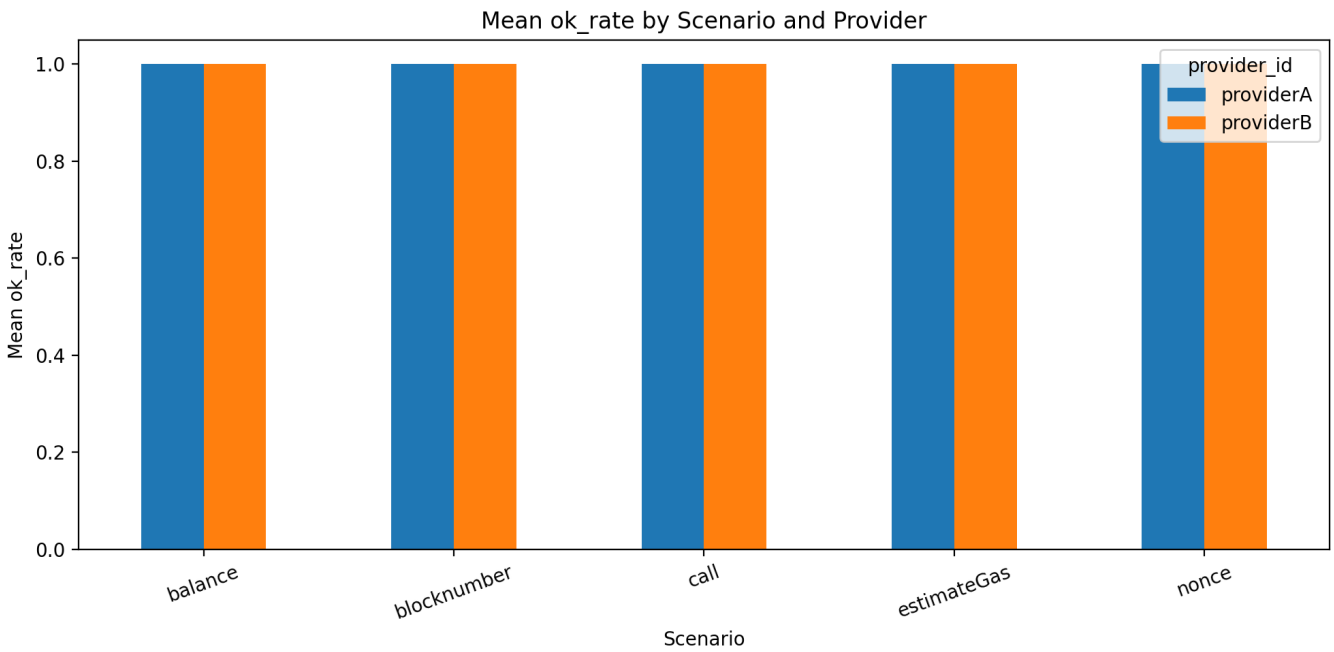
Screenshot 3 — Final p95 latency comparison

What this shows: the figure `results/figures/latency_p95.png` highlights tail-latency differences between providers and scenarios, which is more informative than average latency alone for RPC benchmarking.



Screenshot 4 — Final availability comparison

What this shows: the figure `results/figures/ok_rate.png` confirms that all final experiment runs in the current matrix completed successfully, with `ok_rate = 1.0` across all scenarios and both providers.



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3) Validation I performed

What I validated

- **Final end-to-end reproducibility**:
  - The project can execute the full experiment matrix from configs, produce request-level JSONL logs, summarize results into a single CSV, and generate final figures from that summary.
- **Metric integrity**:

    - `ok_rate`, `median_latency_ms`, `p95_latency_ms`, and `has_address_ratio` are all produced consistently and can be traced back to the request-level logs.
- **Scenario semantics**:
    - `blocknumber` functions as the intended address-free control scenario.
    - `balance`, `nonce`, `call`, and `estimateGas` are represented as address-bearing scenarios under the current parameter construction logic.
- **Provider comparison capability**:
    - The final pipeline supports side-by-side comparison of provider A and provider B across all configured scenarios.

How I validated it

- Executed the final experiment matrix with:

  python src/batch_run.py "configs/m2/*.yaml" results/summary.csv

- Generated final comparison figures with:

  python src/plots.py results/summary.csv results/figures

- Verified that:

    - a new `results/summary.csv` was created for the batch run,
    - all configured runs contributed rows to the final CSV,
    - all four figures were generated successfully under `results/figures/`,
    - the plotted `has_address_ratio` values matched the scenario design,
    - the plotted `ok_rate` values matched the successful batch run.

Result

- The Milestone 3 pipeline completed successfully and produced a **final, report-ready output set** consisting of:
    - raw per-run logs,
    - a consolidated summary table,
    - and automated comparison figures.
- The final figures show that:
    - `blocknumber` remains an address-free control (`has_address_ratio = 0`),
    - the other configured scenarios are address-bearing under the current workload definitions,
    - all runs completed successfully (`ok_rate = 1.0`),
    - and provider-level latency differences are visible in both median and p95 comparisons.
- Compared with Milestone 2, the main improvement is not a new experiment matrix, but a **stronger analysis and delivery layer**: cleaner batch execution, richer summarization, and automated visual outputs suitable for final reporting.

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4) AI usage transparency

- **AI tool used:** ChatGPT
- **How I used AI:** improving project structure, refining experiment-pipeline code, generating plotting and summarization helpers, and improving the wording and organization of final documentation.

- **One AI output I rejected (and why it was wrong, risky, or insufficient):**
  Rejected the suggestion to use **mean latency alone** as the primary performance comparison metric. In RPC measurements, latency distributions can be heavy-tailed, so averages may hide reliability differences. For the final version, I retained and emphasized **median latency, p95 latency, and success rate** as more robust indicators.

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final reproduction commands

- Run the full final experiment matrix:

  ```
  python src/batch_run.py "configs/m2/*.yaml" results/summary.csv
  ```

- Generate final figures:

  ```
  python src/plots.py results/summary.csv results/figures
  ```